

BRIGETTE VAZQUEZ SEGOVIA

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EDUCATION

University of Michigan

Ph.D., Astronomy and Astrophysics, expected 2029

August 2024–Present

Ann Arbor, MI

University of California, San Diego

B.S., Physics with Specialization in Astrophysics; Minor in Mathematics

September 2020–June 2024

San Diego, CA

RESEARCH EXPERIENCE

University of Michigan

Graduate Student Researcher *with Professor Monica Valluri*

August 2024–Present

- Develop modified particle-spray simulations to study how the time-evolving tidal field, mass, inner density profile, and orbit of a parent dwarf-galaxy dark matter halo shape the morphology and kinematics of globular-cluster stellar streams, including gaps, spurs, and tidal heating.

Harvard Center for Astrophysics

Astronomy Summer Intern *with Dr. Priyanka Chakraborty and Dr. Adam Foster*

June 2023–September 2023

- Analyzed Chandra LETG observations of the active galactic nucleus NGC 3227 using XSPEC and Cloudy spectral-synthesis models to constrain the photoionization parameter and column density.

Carnegie Observatories CASSI REU

with Dr. Ana Bonaca

June 2022–June 2024

- Used simulated Milky Way-like systems to investigate how cold dark matter subhalo encounters perturb stellar streams, reconstructing subhalo orbital histories and quantifying their effects on stream kinematics.

University of California, San Diego

with Professor Adam Burgasser

June 2021–June 2022

- Characterized Gaia eDR3 wide binary systems with late-M and L dwarf companions using multi-wavelength survey data and SpeX near-infrared spectral reductions.

PUBLICATIONS

Zhou, T., Jacobsen, D., **Vazquez Segovia, B.**, et al. “Resolved Binaries with Late-M and L Dwarf Companions Identified in Gaia eDR3.” 2023, *Research Notes of the AAS*, 7, 50.

Segovia, B. V., Bonaca, A., & Nadler, E. “Detecting Dark Matter Subhalos Using Gaia-1 and Other Stellar Streams.” 2023, *Bulletin of the AAS*, 55, 2.

SCIENCE PRESENTATIONS

Carnegie Mellon University, Pittsburgh, PA

Accreted Globular Cluster Stellar Stream Microstructures Resulting from Tides from the Parent Dwarf Galaxy

June 2024

243rd American Astronomical Society Meeting, New Orleans, LA

Constraining the Physical Conditions in Active Galactic Nuclei: A Case Study of NGC 3227 [Poster]

January 2024

Harvard-SAO Astrophysics REU Symposium, Cambridge, MA

Constraining the Physical Conditions in Active Galactic Nuclei: A Case Study of NGC 3227

August 2023

241st American Astronomical Society Meeting, Seattle, WA

Detecting Dark Matter Subhalos Using Gaia-1 and Other Stellar Streams [Poster]

January 2023

Carnegie Observatories REU Symposium, Pasadena, CA
Detecting Dark Matter Subhalos Using Gaia-1 and Other Stellar Streams

August 2022

UC San Diego Lightning Talk, San Diego, CA
Curating a Sample of M and L Dwarfs in Wide Binary Systems

February 2022

AWARDS & RECOGNITION

2024–Present **Rackham Merit Fellow**, University of Michigan
2023 **Harvard Astronomy Summer Intern Program**, Harvard Center for Astrophysics
2022 **Carnegie CASSI REU Internship**, Carnegie Observatories
2021 **Triton Research Scholar**, University of California, San Diego
2020–2024 **Provost Honors**, University of California, San Diego
2020–2024 **Morgan Family Foundation Scholarship**
2019 **STEAM Scholarship**
2019 **State of California Senate Certificate of Recognition**
2019 **House of Representatives Certificate of Recognition**
2019 **County of San Diego Certificate of Recognition**

TEACHING

UC San Diego Supplemental Instruction Leader

September 2022–June 2024

- Led weekly study sessions for undergraduate physics and programming courses.
- Facilitated collaborative learning, developed Python coursework, and supported students in strengthening problem-solving skills.
- Coordinated with faculty and conducted review sessions before exams.

OUTREACH & SERVICE

Undergraduate Research Advising
Graduate Student Advisor

Fall 2024–Present

- Support an undergraduate student through monthly advising meetings focused on research progress, coursework, internship applications, and professional development.

University of Michigan Astronomy FEMMES
Coordinator

August 2024–Present

- Coordinate departmental outreach efforts for FEMMES, developing astronomy activities that introduce K–12 students to scientific inquiry, observational thinking, and careers in STEM.
- Design accessible presentations and hands-on demonstrations that connect astronomical concepts to students' everyday experiences.

Senior Outreach Center
Student Volunteer

February 2024–Present

- Collaborate with astronomy graduate students to create community-centered astronomy programming for senior citizens in Ann Arbor.
- Develop presentations and interactive activities that make current topics in astronomy accessible to lifelong learners.

UC San Diego Astronomy Club
Vice President

January 2023–June 2024

- Helped reestablish the UC San Diego Astronomy Club as an active student organization by organizing stargazing nights, outreach events, and community-building activities.
- Led outreach planning, media coordination, and faculty communication to strengthen student engage-

ment with astronomy.

UC San Diego Society of Physics Students

September 2020–June 2024

External Secretary and Outreach Member

- Supported programming and outreach efforts aimed at building an inclusive physics community for undergraduate students.
- Organized events, maintained communication calendars, and helped coordinate faculty talks and student-facing academic resources.

SKILLS _____

PROGRAMMING & SCIENTIFIC COMPUTING: Python, MATLAB, IDL

TECHNICAL SKILLS: Numerical simulation, data analysis, scientific visualization, and computational modeling

LANGUAGES: Native fluency in English and Spanish; beginner French